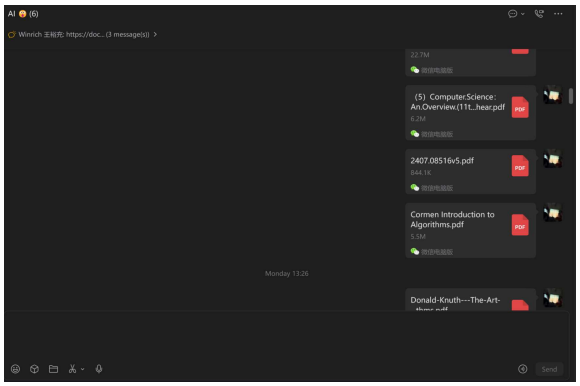
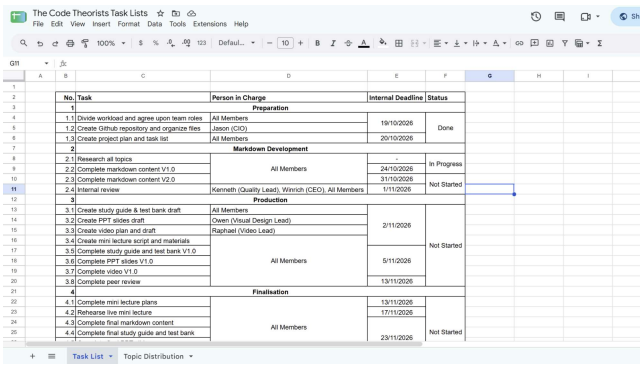
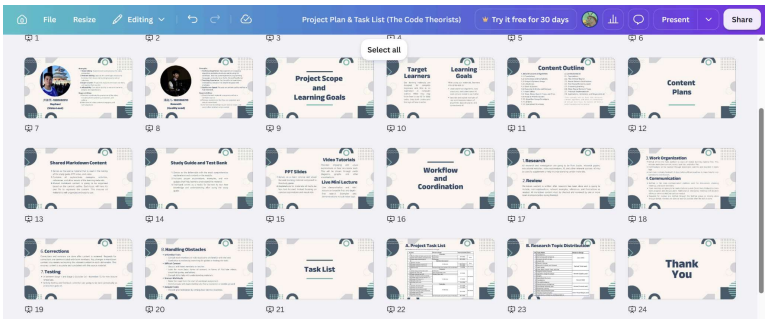
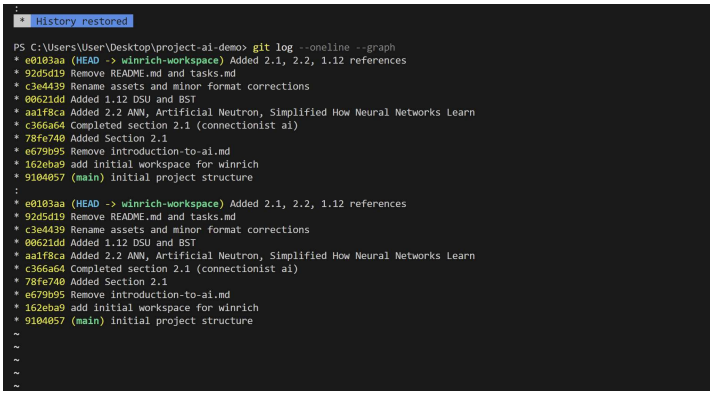
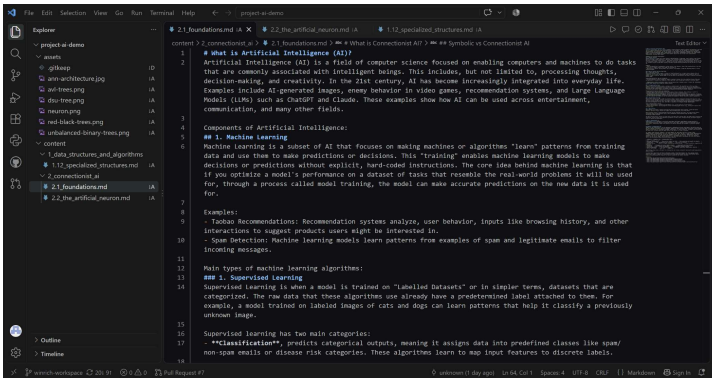
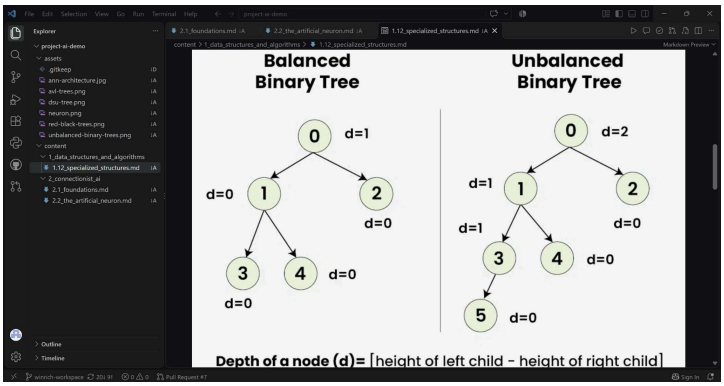
Personal Learning and Work Log
The Code Theorists
Stage I
(As of 24/10/2026)

1. Winrich 1820262257 王裕充 (CEO)

Date	Details
20/10/2026	**Activity:** I did research about the fundamentals of AI, including its definition, machine learning, deep learning, and other aspects. During this period, I also collected materials and other references to aid with research. **Contribution:** I collected useful materials, like books, papers, and links to aid with research. **Outcome and Evidence:** I gained a better understanding on implementations of AI and the basics of connectionist AI.  **Learning and Next:** I found that having a better understanding in the foundations makes it much easier to understand the later topics.
21/10/2026	**Activity:** Finish project plans and task list **Contribution:** I coordinated most of the planning and led all discussions regarding team tasks. I planned the outline of the team tasks, distributed research topics, set internal deadlines, and moved everything to PPT slides. **Outcome and Evidence:** Fully finished project plan and task list.

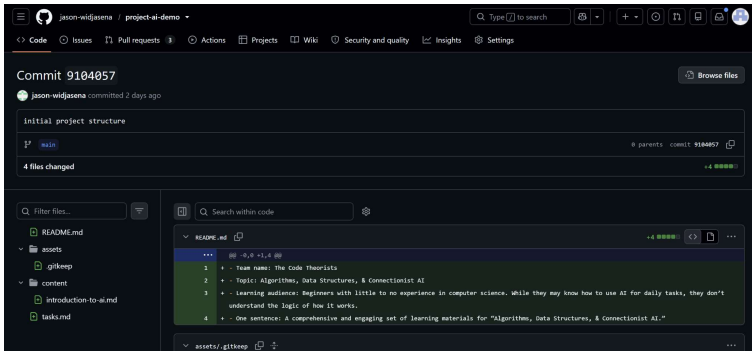
	 The Code Theorists Task Lists File Edit View Insert Format Data Tools Extensions Help Task List Topic Distribution  File Resize Editing Project Plan & Task List (The Code Theorists) Try it free for 30 days Present Share Select all Project Scope and Learning Goals Target Learners Learning Goals Content Outline Content Plans Workflow and Coordination Task List Thank You **Learning and Next:** I gained a clearer understanding of how to lead my team and manage work so that deadlines are met. Before this, I felt stressed about how the team is going to meet the deadlines on time because I didn't have a clear idea of the team workflow and what needs to be done. Next,
22/10/2026	**Activity:** I fully researched about the foundations of AI and connectionist AI, the artificial neuron, and specialized data structures. Then, created the markdown content. **Contribution:** I researched markdown content material mostly through cross-checking websites and additional explanations from AI. I mostly found my information from GeeksforGeeks, W3Schools, and IBM. **Outcome and Evidence:** Finished draft of markdown content 1.12, 2.1, and 2.2  History restored PS C:\Users\User\Desktop\project-ai-demo> git log --oneline --graph * e0103aa (HEAD -> winrich-workspace) Added 2.1, 2.2, 1.12 references * 92d5d19 Remove README.md and tasks.md * c3e4439 Rename assets and minor format corrections * 00621dd Added 1.12 DSU and BST * aa1f8ca Added 2.2 ANN, Artificial Neutron, Simplified How Neural Networks Learn * c366a64 Completed section 2.1 (connectionist ai) * 78fe740 Added Section 2.1 * e679b95 Remove introduction-to-ai.md * 162eba9 add initial workspace for winrich * 9104057 (main) initial project structure * e0103aa (HEAD -> winrich-workspace) Added 2.1, 2.2, 1.12 references * 92d5d19 Remove README.md and tasks.md * c3e4439 Rename assets and minor format corrections * 00621dd Added 1.12 DSU and BST * aa1f8ca Added 2.2 ANN, Artificial Neutron, Simplified How Neural Networks Learn * c366a64 Completed section 2.1 (connectionist ai) * 78fe740 Added Section 2.1 * e679b95 Remove introduction-to-ai.md * 162eba9 add initial workspace for winrich * 9104057 (main) initial project structure



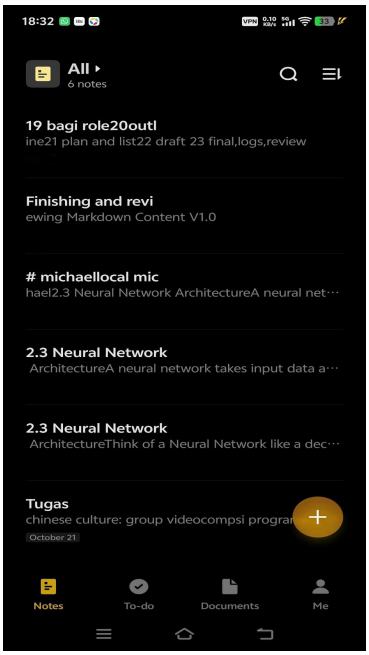


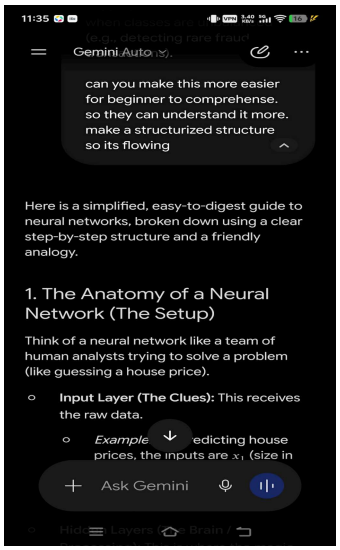
Learning and Next: I learned the main points of the three materials and how to explain it in simpler methods for beginners. Next, I want to add more in depth mathematical explanations and relatable examples so it's easier to understand.

2. Jason 1820262246 庄翔俊 (CIO)

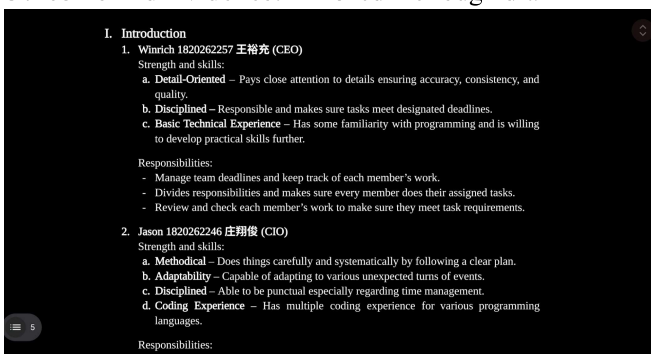
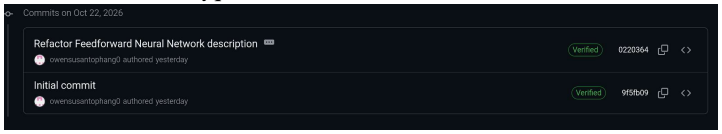
Date	Details
22/10/2026	**Activity:** I used GitHub to create the repository which later on I connected it with VS Code. In VS Code, I created the initial project structure and the initial workspace for working later on. **Contribution:** Created initial repository and the workspace for my team, the initial project structure, and the initial my personal tasks-workspace. **Outcome and Evidence:** I finished creating the initial setup for the group to later work on  **Learning and Next:** Learn how to create the repository and also how to connect and use the VS Code.
23/10/2026	**Activity:** I used GitHub, VS Code, and Gemini AI to help me finish my tasks, including adding my content to GitHub from VS Code where my content is checked by Gemini AI for its validity. I also used Gemini AI to help me fix the problems surrounding GitHub. **Contribution:** Restructure the repository, finish up my part of group tasks, and fix the issues in the repository. **Outcome and Evidence:** I finished 1.1 to 1.4 which is my part complete with the explanation and visual for some subcategory. **Learning and Next:** Learn about my material more so I am more knowledgeable about algorithms and data structures.

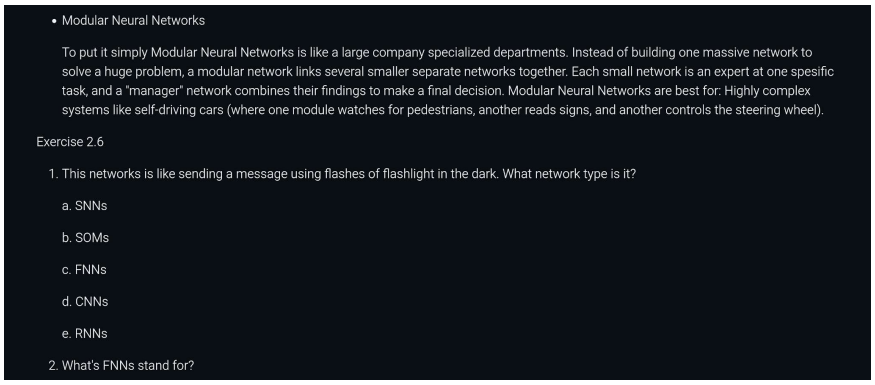
3. Michael 1820262227 张天龙 (Content Lead)

Date	Details
22/10/2026	**Activity:** I use Gemini AI to scan the reference book I got from my senior. I asked which topics are related to our group's theme. And so, I began crosschecking from the answer and planned the text on my notes. **Contribution:** I wrote the initial draft in my local GitHub repository, rephrased and deleted unnecessary words to make it easier for beginners. **Outcome and Evidence:** Finished initial draft  (Screenshot of the draft in my notes) **Learning and Next:** Learning the basic materials and completing my draft
23/10/2026	**Activity:** Completing the draft, adding questions for the reader, paraphrase for better readability, submitting to GitHub branch, adding a bunch of information on project logs and meeting summaries **Contribution:** Aside from submitting the topic into GitHub, I think my main contribution is helping around the reviewing part to correct small errors in topic 1 and fulfilling project logs and meeting summaries. **Outcome and Evidence:** The outcome is supposed to be the finalization draft on GitHub and the completion of meeting summaries and project logs.

	 (Screenshot of my prompt to Gemini to paraphrase my initial draft) A lot of the answers got rejected by me, so I revised a lot. The result is that the text that I submitted is easier to comprehend for beginners, contains examples, and relevant activities. **Learning and Next:** Understanding more complex definition and neural use in real life, more aware of such errors and focused on analysis, and merge all files in GitHub.

4. Owen 1820262252 彭世涛 (Visual Design Lead)

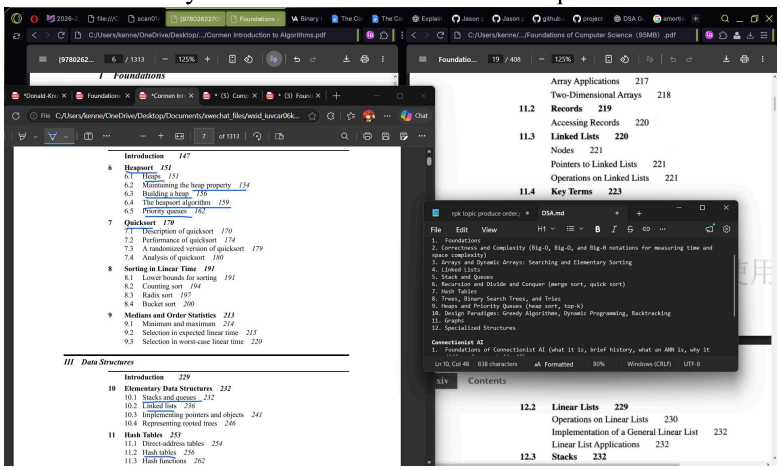
Date	Details
19/10/2026	**Activity:** Started the rough draft of the project plan and task list **Contribution:** I worked on filling in the coordination part. **Outcome and Evidence:** Finished the rough draft  I. Introduction 1. Winrich 1820262257 王裕充 (CEO) Strength and skills: a. Detail-Oriented – Pays close attention to details ensuring accuracy, consistency, and quality. b. Disciplined – Responsible and makes sure tasks meet designated deadlines. c. Basic Technical Experience – Has some familiarity with programming and is willing to develop practical skills further. Responsibilities: - Manage team deadlines and keep track of each member's work. - Divides responsibilities and makes sure every member does their assigned tasks. - Review and check each member's work to make sure they meet task requirements. 2. Jason 1820262246 庄翔俊 (CIO) Strength and skills: a. Methodical – Does things carefully and systematically by following a clear plan. b. Adaptability – Capable of adapting to various unexpected turns of events. c. Disciplined – Able to be punctual especially regarding time management. d. Coding Experience – Has multiple coding experience for various programming languages. Responsibilities: **Learning and Next:** Learned how to solve problems and the plans throughout the project. Research about my designated topic.
22/10/2026	**Activity:** I started researching on my designated topic which is 2.6 Major Neural Network Types. With the help of Gemini AI I started to understand the topic more and more. **Contribution:** Wrote the initial draft to my notes app then moved it into my local GitHub. While also cross checking whether the information given by Gemini AI is correct using sources given by it. **Outcome and Evidence:** I began to write about FFNs (Feedforward Neural Networks) the first major neural network i put, and also an introduction of Major Neural Network Type.  Commits on Oct 22, 2026 Refactor Feedforward Neural Network description owensusantophang0 authored yesterday Verified 0220364 Initial commit owensusantophang0 authored yesterday Verified 9f5fb09 I began writing my initial draft on 22 October. **Learning and Next:** Learned about Major Neural Network types. Finishing my Draft and moving it to the main project GitHub.
23/10/2026	**Activity:** Finishing my draft on my notes app and moving it to the main project GitHub. Add a 2.6 Exercise. **Contribution:** Wrote the whole Major Neural Network Types, and made my own questions. **Outcome and Evidence:** The 2.6 subtopic are finished.

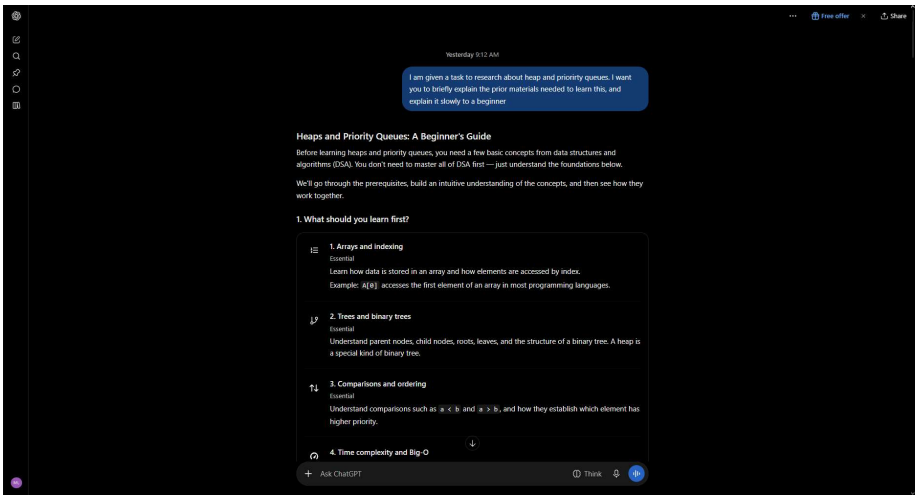
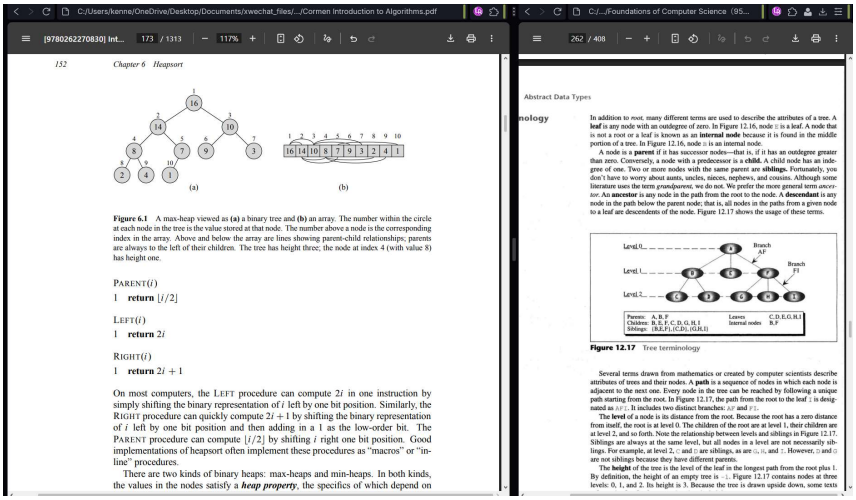
	 2.6 Exercise, and the finished draft of Major Neural Network Types topic. **Learning and Next:** Learn more on Major Neural Networks and how to make the title and subtitle on GitHub. Started researching on 2.7 Practical Implementation of Connectionist AI.

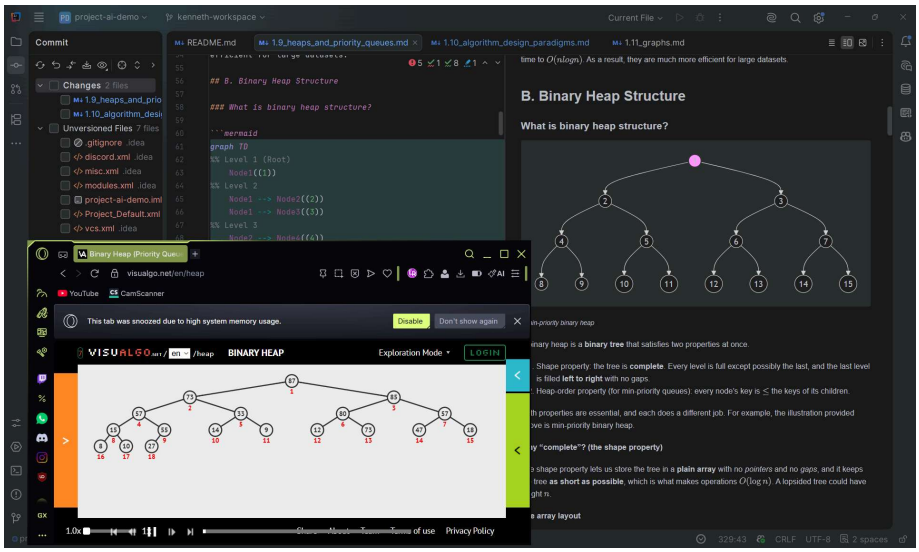
5. Raphael 1820262212 洪建民 (Video Lead)

Date	Details
22/10/2026	**Activity**: I read CLRS §10.1 (topic 1.5), and CLRS §2.3.1 and Brookshear's chapter on recursive structures (topic 1.6), then used an AI tool to simplify the explanations for beginners. I rechecked every table, calculation, visual, and answer by hand. **Contribution**: I wrote the drafts of topics 1.5 (Stacks & Queues) and 1.6 (Recursion & Divide-and-Conquer) in Google Docs. **Outcome and evidence**: I completed first drafts of 1.5 and 1.6, each with a worked example, a visual, practice questions with answers, and references. **Learning and Next**: Everyday analogies (plates, a cashier line, a guessing game) make abstract ideas easier for beginners. In 1.6 I also learned that recursion needs a base case and that divide-and-conquer adds a combine step.
23/10/2026	**Activity**: I wrote drafts of topics 1.7 (Hash Tables) and 1.8 (Trees, BSTs, and Tries), then converted topics 1.5 to 1.8 into Markdown and pushed them to the team's GitHub repository. **Contribution**: I studied CLRS Chapters 11 and 12, Sedgewick on tries, and two GeeksforGeeks articles, and used an AI tool to simplify them. I checked every calculation, table, and answer manually, and did the Markdown conversion and uploaded it myself. **Outcome and evidence**: First drafts of 1.7 and 1.8, and Markdown files for 1.5 to 1.8 pushed to branch. **Learning and Next**: An analogy helps beginners, but each topic also needs a worked example and a visual to show how the data is stored. Next, I will open a pull request, ask the team to review it together, fix the necessary things, and merge into main.

6. Kenneth 1820262213 潘超凡 (Quality Lead)

Date	Details
19/10/2026	**Activity:** I read through some books' outlines regarding algorithm and data structure and made a sketch/outline regarding the outline of the study guide, to better understand and divide our workload. **Contribution**: I sketched the subtopics we are going to cover in this course and asked for AI's input on it, and how deep we should cover every sub-topic, to better prepare my team members for their sub-topics. **Outcome and Evidence**: I wrote all the subtopics in the project plan, and researched briefly about each of them and their importance.  **Learning and Next**: Originally, I wanted to split the study guide into three parts: Algorithm, Data Structures and Connectionist AI, but whilst researching I learned that Data Structures and Algorithm are very closely connected together, and decided to combine them together, where we first teach the underlying data structures relevant, then teach about the algorithms related to it. Next up, I need to research deeply about the topic assigned to me.
22/10/2026	**Activity:** I researched all the topics (Heaps and Priority Queues, Algorithm and Design Paradigm, Graphs) assigned to me briefly, and divided them into subtopics I should cover. **Contribution**: I read some books regarding the topic, and researched online, and with the help of AI to determine what I need to cover and how deeply I need to cover them. Also, for myself, I also enforced my own learning about the GitHub workflow. **Outcome and Evidence**: I sketched the outline of my subtopics, and briefly introduced what I wanted to cover in each of them.

	  **Learning and Next**: I read pretty deeply about Heaps and Priority Queues in this period, and I now know about how they work, how they are completed binary trees, what inserting and extraction mean and how they work. I also touched briefly on how the build heap problem arises, the solution and how heapsort works ("keep extracting and place it in the end of the array") and some usage patterns. Next up I need to try to create a sample of the study guide and use my knowledge and skill to simplify it.
23/10/2026	**Activity:** Researched all the topics assigned to me deeply, and created a sample study guide. **Contribution**: I researched online, and via some applications such as visualgo.net to deepen my understanding, then created a sample study guide on the first topic I was assigned to. **Outcome and evidence**: I tried to put together the explanations in detail and thoroughly, and tried to make it understandable for beginners by making analogies and such.



Learning and Next: I found that trying to write my own guide helps me actually help me understand it much better than before, and it helps to put my thoughts into a more structured space. Next up, I need to continue on with the study guide.